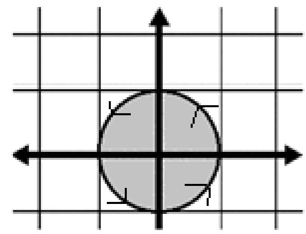
MTH 202 Chapter 5: Topics In Vector Calculus

5.4 Green's Theorem Worksheet

1. Consider the work done by the vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$ on a particle moving counterclockwise around the unit circle.
   a) Use Green's Theorem to evaluate the resulting line integral.

   A picture of the curve C and implied region D are shown at the right.

   Here, $D = \{(r,\theta) | 0 \le \theta \le 2\pi \ \& \ 0 \le r \le 1\}$.

   We note that $P(x,y) = y \ \& \ Q(x,y) = -x$ and so $\frac{\partial P}{\partial y} = 1 \ \& \ \frac{\partial Q}{\partial x} = -1$.

   We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy-plane and that P and Q have continuous first order partial derivatives on D.

   Therefore, by Green's Theorem,

   $$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C y\,dx - x\,dy = \iint_D (-1-1)\,dA = \iint_D -2\,dA = -2\iint_D dA = -2\pi$$

   b) How would your answer change if the particle was moving clockwise around the unit circle?

   Moving in the clockwise direction would change our path C to $-C$. Thus, the only difference would be that the total work would be multiplied by negative one.

   $$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r} = -\int_C y\,dx - x\,dy = -\iint_D (-1-1)\,dA = \iint_D -2\,dA = 2\iint_D dA = 2\pi.$$

2. Use Green's Theorem to evaluate the line integral where $\mathbf{F} = xy\mathbf{i} + (x-y)\mathbf{j}$ and C is the triangle joining $(2,0)$, $(0,2)$, and $(-2,0)$ in the counterclockwise direction.

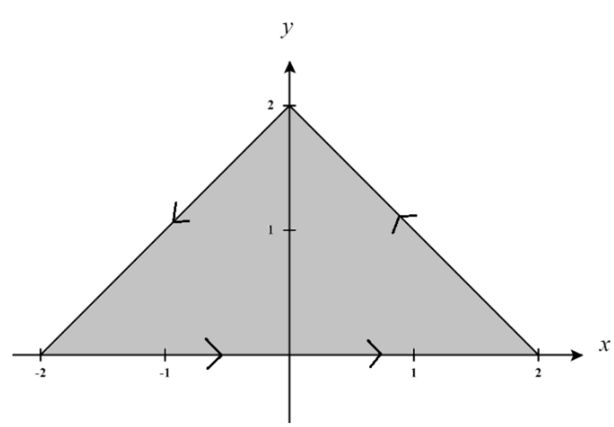
   A picture of the curve C and implied region D are shown at the right.

   Here, $D = \{(x,y) | y-2 \le x \le 2-y \ \& \ 0 \le y \le 2\}$.

   We note that $P(x,y) = xy \ \& \ Q(x,y) = x - y$ and so

   $\frac{\partial P}{\partial y} = x \ \& \ \frac{\partial Q}{\partial x} = 1$.

   We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy-plane and that P and Q have continuous first order partial derivatives on D.

Therefore, by Green's Theorem,

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C xy\,dx + (x-y)\,dy = \iint_D (1-x)\,dA = \int_0^2 \int_{y-2}^{2-y} (1-x)\,dA = \int_0^2 \left(x - \frac{1}{2}x^2 \right)\Bigg|_{y-2}^{2-y}$$

$$= \int_0^2 \left[\left((2-y) - \frac{1}{2}(2-y)^2 \right) - \left((y-2) - \frac{1}{2}(y-2)^2 \right) \right] dy$$

$$= \int_0^2 \left[2-y-2+2y-\frac{1}{2}y^2 - \left(y-2-\frac{1}{2}y^2+2y-2 \right) \right] dy$$

$$= \int_0^2 \left[2-y-2+2y-\frac{1}{2}y^2 - y+2+\frac{1}{2}y^2-2y+2 \right] dy$$

$$= \int_0^2 [4-2y]\,dy$$

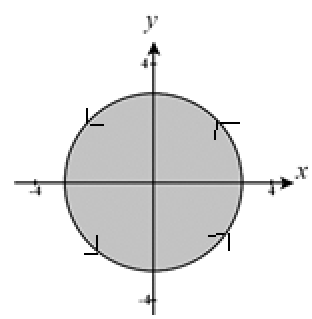
$$= \left(4y-y^2\right)\Big|_0^2$$

$$= (8-4)+(0)$$

$$= 4$$

3. Evaluate $\oint_C \left(x^2+y^2\right)dx + x\,dy$, where C is the circle $x^2+y^2=9$.

A picture of the curve C and implied region D are shown at the right.



Here, $D = \left\{ (r,\theta) \mid 0 \le \theta \le 2\pi \ \ \& \ \ 0 \le r \le 3 \right\}$.

We note that $P(x,y) = x^2+y^2$ & $Q(x,y) = x$ and so $\dfrac{\partial P}{\partial y} = 2y$ & $\dfrac{\partial Q}{\partial x} = 1$.

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy-plane and that P and Q have continuous first order partial derivatives on D.

Therefore, by Green's Theorem,

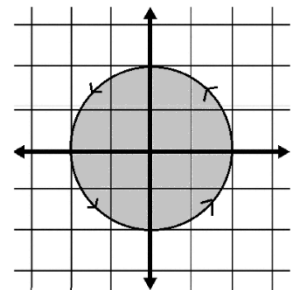
$$\oint_C \left(x^2+y^2\right)dx + x\,dy = \iint_D (1-2y)\,dA = \int_0^{2\pi}\int_0^3 \left(1-2r\sin(\theta)\right) r\,dr\,d\theta = \int_0^{2\pi}\int_0^3 \left(r-2r^2\sin(\theta)\right) dr\,d\theta$$

$$= \int_0^{2\pi} \left[\frac{1}{2}r^2 - \frac{2}{3}r^3\sin(\theta) \right]\Bigg|_0^3 d\theta = \int_0^{2\pi} \left[\frac{9}{2} - 18\sin(\theta) \right] d\theta$$

$$= \left[\frac{9}{2}\theta + 18\cos(\theta) \right]\Bigg|_0^{2\pi} = [9\pi+18] - [0+18]$$

$$= 9\pi$$

4. Suppose an object is being spun by high wind currents modelled by the vector field $\mathbf{F}(x,y)=\left\langle -y^3+x\tan^{-1}(y), x^3+2+\dfrac{x^2}{2+2y^2}\right\rangle$ in a counterclockwise fashion around a circle of radius 2. Determine the total work done by the wind currents on the object as the object moves around the circle once.



A picture of the curve C and implied region D are shown at the right.

Here, $D=\{(r,\theta)|0\le\theta\le 2\pi \ \& \ 0\le r\le 2\}$.

We note that $P(x,y)=-y^3+x\tan^{-1}(y)$ & $Q(x,y)=x^3+2+\dfrac{x^2}{2+2y^2}$ and

so $\dfrac{\partial P}{\partial y}=-3y^2+\dfrac{x}{1+y^2}$ & $\dfrac{\partial Q}{\partial x}=3x^2+\dfrac{x}{1+y^2}$.

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy-plane and that P and Q have continuous first order partial derivatives on D.

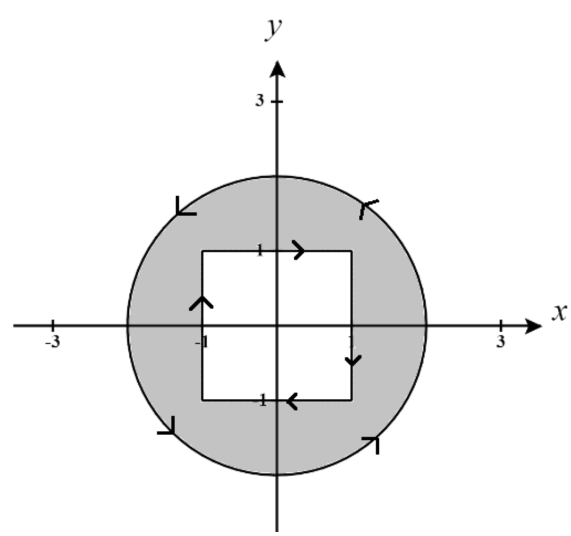
Therefore, by Green's Theorem,

$$\int_C \mathbf{F}\cdot d\mathbf{r}=\int_C\left(-y^3+x\tan^{-1}(y)\right)dx+\left(x^3+2+\frac{x^2}{2+2y^2}\right)dy=\iint_D\left[\left(3x^2+\frac{x}{1+y^2}\right)-\left(-3y^2+\frac{x}{1+y^2}\right)\right]dA$$

$$=\iint_D\left(3x^2+3y^2\right)dA=\int_0^{2\pi}\int_0^2 3r^2\cdot r\,dr\,d\theta$$

$$=\int_0^{2\pi}\frac{3}{4}r^4\Big|_0^2\,d\theta=\int_0^{2\pi}12d\theta=24\pi$$

5. Determine the work done by the vector field $\mathbf{F}(x,y)=\left\langle e^{\cosh(x)}+3y, e^{\sinh(y)}-3x\right\rangle$ on a particle moving along C, where C is the boundary of the region outside the unit square but inside the circle $4x^2+4y^2=16$ with positive orientation.
A picture of the curve C and implied region D are shown at the right.



We note that $P(x,y)=e^{\cosh(x)}+3y$ & $Q(x,y)=e^{\sinh(y)}+3x$

and so $\dfrac{\partial P}{\partial y}=3$ & $\dfrac{\partial Q}{\partial x}=-3$.

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy-plane and that P and Q have continuous first order partial derivatives on D.

Therefore, by Green's Theorem,

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \left(e^{\cosh(x)} + 3y\right)dx + \left(e^{\sinh(y)} - 3x\right)dy = \iint_D (-3-3)\,dA = \iint_D -6dA = -6\iint_D dA$$

We know from basic geometry that the area of the region is given by $A = \pi(2)^2 - 1 = 4\pi - 1$

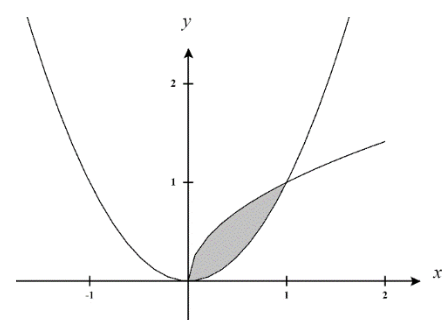
Thus, $\int_C \mathbf{F}\cdot d\mathbf{r} = -6\iint_D dA = 6 - 24\pi$

6. Use Green's Theorem to evaluate the line integral along the given positively oriented curve.

$$\int_C \left(2x + \cos\left(y^2\right)\right)dy + \left(y + e^{\sqrt{x}}\right)dx$$

where C is the boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.

A picture of the curve C and implied region D are shown at the right.



Here, $D = \left\{(x,y) \mid x^2 \le y \le \sqrt{x} \;\;\&\;\; 0 \le x \le 1\right\}$.

We note that $P(x,y) = y + e^{\sqrt{x}}$ & $Q(x,y) = 2x + \cos\left(y^2\right)$ and so $\dfrac{\partial P}{\partial y} = 1$ & $\dfrac{\partial Q}{\partial x} = 2$.

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the *xy*-plane and that P and Q have continuous first order partial derivatives on D.

Therefore, by Green's Theorem,

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \left(2x + \cos\left(y^2\right)\right)dy + \left(y + e^{\sqrt{x}}\right)dx = \iint_D (2-1)\,dA = \iint_D dA$$

$$= \int_0^1 \int_{x^2}^{\sqrt{x}} dy\,dx = \int_0^1 (y)\Big|_{x^2}^{\sqrt{x}} dx$$

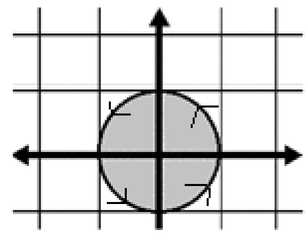
$$= \int_0^1 \left(\sqrt{x} - x^2\right)dx = \left(\frac{2}{3}x^{3/2} - \frac{1}{3}x^3\right)\Bigg|_0^1$$

$$= \left(\frac{2}{3} - \frac{1}{3}\right) = \frac{1}{3}$$

7. Explain why green's theorem cannot be used to evaluate the following line integral:

$$\int_C \ln(1+2y)\,dx + \frac{3}{2}\left(x-\frac{1}{2}\right)^{2/3} dy$$ where C is the boundary of positively oriented unit circle.

A picture of the curve C and implied region D are shown at the right.

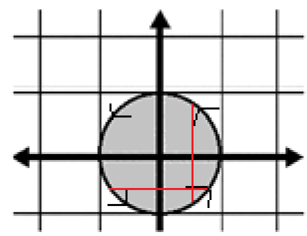


Here, $D=\{(r,\theta)|0\le\theta\le 2\pi \;\&\; 0\le r\le 1\}$.

We note that $P(x,y)=\ln(1+2y)$ & $Q(x,y)=\frac{3}{2}\left(x-\frac{1}{2}\right)^{2/3}$ and so

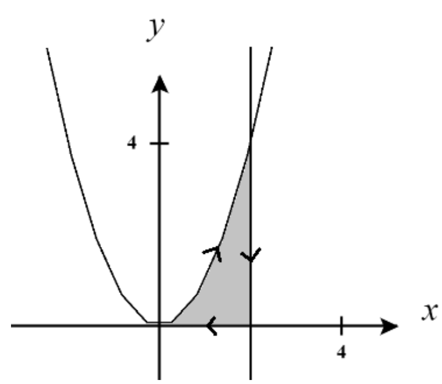
$$\frac{\partial P}{\partial y}=\frac{2}{1+2y} \;\&\; \frac{\partial Q}{\partial x}=\frac{1}{\sqrt[3]{x-\frac{1}{2}}}.$$

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the *xy*-plane. However, we see that $\frac{\partial P}{\partial y}$ is discontinuous when $y=-\frac{1}{2}$ and $\frac{\partial Q}{\partial x}$ is discontinuous when $x=\frac{1}{2}$. Therefore, the first partials are not continuous on D. In fact, the first partials at all points shown in red in the picture to the right.



8. Compute $\int_C xe^{-2x}dx+\left(x^4+2x^2y^2+y^4\right)dy$, where C is the negatively oriented boundary of the region enclosed by $y=x^2$, the *x*-axis, and $x=2$.

A picture of the curve C and implied region D are shown at the right.



Here, $D=\{(x,y)|0\le y\le x^2 \;\&\; 0\le x\le 2\}$.

We note that

$P(x,y)=xe^{-2x}$ & $Q(x,y)=x^4+2x^2y^2+y^4=\left(x^2+y^2\right)^2$ and so

$$\frac{\partial P}{\partial y}=0 \;\&\; \frac{\partial Q}{\partial x}=4x^3+4xy^2=4x\left(x^2+y^2\right).$$

We see that -C is a positively oriented, piecewise-smooth, simple closed curve in the *xy*-plane and that P and Q have continuous first order partial derivatives on D.

Therefore, by Green's Theorem,

$$\int_C \mathbf{F}\cdot d\mathbf{r} = -\int_{-C} \mathbf{F}\cdot d\mathbf{r} = -\int_{-C} \left(xe^{-2x}\right)dx + \left(x^4+2x^2y^2+y^4\right)dy = -\iint_D \left(4x^3+4xy^2\right)dA$$

$$= -\int_0^2\int_0^{x^2} \left(4x^3+4xy^2\right)dy\,dx = -\int_0^2 \left(4x^3y+\frac{4}{3}xy^3\right)\Bigg|_0^{x^2} dx = -\int_0^2 \left(4x^5+\frac{4}{3}x^7\right)dx = -\left(\frac{2}{3}x^6+\frac{1}{6}x^8\right)\Bigg|_0^2$$

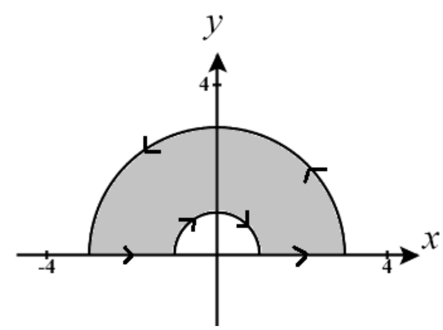
$$= -\left(\frac{2}{3}(64)+\frac{1}{6}(256)\right)$$

$$= -\frac{256}{3}$$

9. Use Green's Theorem to evaluate the following line integral where C is the path from $(1,0)$ to $(3,0)$ along a line, followed by the path from $(3,0)$ to $(-3,0)$ along the top half of a circle of radius 3 centered at the origin, followed by the path from $(-3,0)$ to $(-1,0)$ along a line, followed by the path from $(-1,0)$ to $(1,0)$ along the top half of a circle of radius 1 centered at the origin.

$$\int_C \left(\tan^{-1}(x)+y^2\right)dx + \left(e^y - x^2\right)dy$$

A picture of the curve C and implied region D are shown at the right.
Here, $D = \{(r,\theta) | 0 \le \theta \le \pi \;\&\; 1 \le r \le 3\}$.



We note that $P(x,y) = \tan^{-1}(x)+y^2 \;\&\; Q(x,y) = e^y - x^2$ and so

$$\frac{\partial P}{\partial y} = 2y \;\&\; \frac{\partial Q}{\partial x} = -2x.$$

We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy-plane and that P and Q have continuous first order partial derivatives on D.

Therefore, by Green's Theorem,

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \left(\tan^{-1}(x)+y^2\right)dx + \left(e^y-x^2\right)dy = \iint_D (-2x-2y)\,dA$$

$$= -2\iint_D (x+y)\,dA = -2\int_0^\pi\int_1^3 \left(r\cos(\theta)+r\sin(\theta)\right)\cdot r\,dr\,d\theta$$

$$= -2\int_0^\pi\int_1^3 r^2\left[\cos(\theta)+\sin(\theta)\right]dr\,d\theta = -2\int_0^\pi \frac{1}{3}r^3\left[\cos(\theta)+\sin(\theta)\right]\Bigg|_1^3 d\theta$$

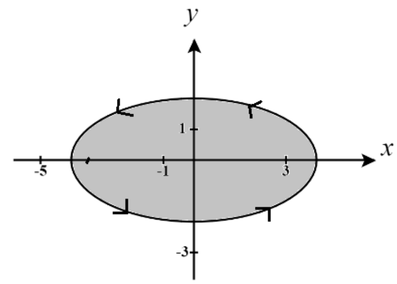
$$= -2\int_0^\pi \frac{26}{3}\left[\cos(\theta)+\sin(\theta)\right]d\theta = -\frac{52}{3}\int_0^\pi \left[\cos(\theta)+\sin(\theta)\right]d\theta$$

$$= -\frac{52}{3}\left[\sin(\theta)-\cos(\theta)\right]\Big|_0^\pi = -\frac{52}{3}\left[(0+1)-(0-1)\right] = -\frac{52}{3}[2]$$

$$= -\frac{104}{3}$$

10. Evaluate $\oint_C \left(x^{\sin(x)} - x - 2y\right)dx + \left(2x - \tan^{-1}(y) - y^{\cos(y)}\right)dy$, where C is the ellipse $x^2 + 4y^2 = 16$.

A picture of the curve C and implied region D are shown at the right.

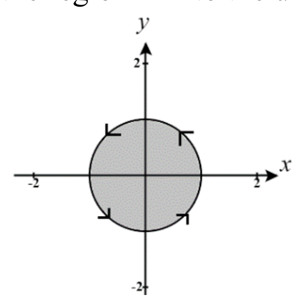


We note that
$P(x,y) = x^{\sin(x)} - x - 2y$ & $Q(x,y) = 2x - \tan^{-1}(y) - y^{\cos(y)}$ and so
$\dfrac{\partial P}{\partial y} = -2$ & $\dfrac{\partial Q}{\partial x} = 2$.
We see that C is a positively oriented, piecewise-smooth, simple closed curve in the xy-plane and that P and Q have continuous first order partial derivatives on D.

Therefore, by Green's Theorem,

$$\oint_C \left(x^{\sin(x)} - x - 2y\right)dx + \left(2x - \tan^{-1}(y) - y^{\cos(y)}\right)dy = \iint_D \left(2-(-2)\right)dA = \iint_D 4dA = 4\iint_D dA$$

To calculate this more easily we will use a change of variables to rework the region D into the unit circle.



Let $x = 4u$ & $y = 2v$ then we see that S is determined by

$$x^2 + 4y^2 = 16 \Rightarrow (4u)^2 + 4(2v)^2 = 16$$
$$\Rightarrow 16u^2 + 16v^2 = 16 \Rightarrow u^2 + v^2 = 1$$

So, $S = \{(r,\theta) \mid 0 \le \theta \le 2\pi \text{ \& } 0 \le r \le 1\}$ and has the following picture

Therefore, our transformation is $T(u,v) = (4u, 2v)$ and the Jacobian of this transformation is

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} = 8 - 0 = 8$$

Thus,

$$\begin{aligned}\oint_C \left(x^{\sin(x)} - x - 2y\right)dx + \left(2x - \tan^{-1}(y) - y^{\cos(y)}\right)dy &= 4\iint_D dA \\ &= 4\iint_S |8|\,du\,dv \\ &= 32\int_0^{2\pi}\int_0^1 r\,dr\,d\theta \\ &= 32\int_0^{2\pi} \frac{1}{2}d\theta \\ &= 32\pi\end{aligned}$$

NOTE: You may notice that in general the area of an ellipse is given by $A = ab\pi$ where a & b are the radii in each direction of the ellipse. In the example above we have $A = (4)(2)\pi$ and thus

$$4\iint_D dA = 4 \cdot 8\pi = 32\pi$$